Energy Causes Change

Chapter 1

You walk into a dark room. You flip a switch. What changes? The room changes from dark to light.

You are holding a balloon in a quiet place. You stick a pin into the balloon. What changes? The silence in the room changes when you hear a loud sound—*POP*!

A toy car is sitting on top of a table. You give the car a push. What changes? The toy car changes from sitting still to moving.

Whenever there is a **change**, there is some form of **energy** causing the change. Energy is the ability to cause change.

Energy exists in many different forms. It causes different types of changes.

Big Question

Where can we observe evidence of energy causing change?

Vocabulary

change, v. to become different

energy, n. the ability to cause change



Popping a water balloon causes a noise and a splash!

Change Is Evidence of Energy

When you see changes in the world around you, you are seeing evidence, or proof, of energy. Look at the picture. What changes do you think are happening?

As the wave crashes onto the shore, the water moves. That is one change. Imagine you are there on the seashore. You hear the sound of the water splashing against sand and rocks. The sound changes because of the movement of the water and the wind.

Another change is the movement of the sand as it is pushed by the water. And there's a change happening that you can't see—the temperature of the sand is changing as it is warmed by the bright sunlight or cooled by the water.

All these changes are caused by some form of energy.



All of the changes that happen on this seashore are the result of energy.

Energy Occurs in Different Forms

Energy exists in many forms. For example, think of a carousel—a merry-go-round that you might see at a carnival or amusement park.

Motion: The carousel spins around. The horses move riders up and down. All these moving objects have energy of motion.



You can observe evidence of many forms of energy in a spinning carousel.

Sound: As the carousel spins around, it plays loud, cheerful music. The riders laugh and shout to friends. The carousel and the riders are producing sound, another form of energy.

Light: The bright bulbs on the carousel light up the ride. The bulbs give off the form of energy called light.

Heat: If you're riding a carousel, don't touch the bright light bulbs. Why? Because they are hot! The bright bulbs not only produce light, but also give off heat, another form of energy.

Electrical Energy: The carousel, its musical speakers, and its shining lights are all powered by another form of energy. If you look closely around the ride, you will find electrical cords connecting the carousel to a source of electrical energy.

Objects Can Store Energy

When you see the waves crashing on a beach or watch a spinning carousel, you can see changes happening. The changes are evidence of energy. But energy can also be present even when you don't see such evidence.

When a rubber band is not stretched, it isn't causing any change. There is no evidence of energy. But when the rubber band is stretched, it now has the ability to cause changes if it is released, or let go.

The stretched rubber band has **stored energy**. When the rubber band is released, it moves and makes a sound. Those changes are evidence that energy was stored in the stretched rubber band.

Inside a battery are chemicals that store energy. When you put a battery in a flashlight or a toy, the stored chemical energy will be released and cause a change. The energy can make the flashlight light up or the toy move.

Vocabulary

stored energy, n. energy that has the ability to cause change at a later time



What happens when one side of a stretched rubber band is released? *SNAP*!

The Stored Energy of Position

Batteries and stretchy materials can store energy because of the materials of which they are made. Any object also can have stored energy because of its position.

Imagine that you pick up an apple and hold it up above your head. Like a stretched rubber band, the apple has the ability to change position. If you release the apple, it will fall.

When an apple hangs from a tree branch or when you hold the apple above the ground, it has stored energy because of its position. The same is true if you hold up a ball or a book or a bucket—each object has stored energy because it has the ability to cause a change once it starts moving.



This apple has stored energy because of its position.

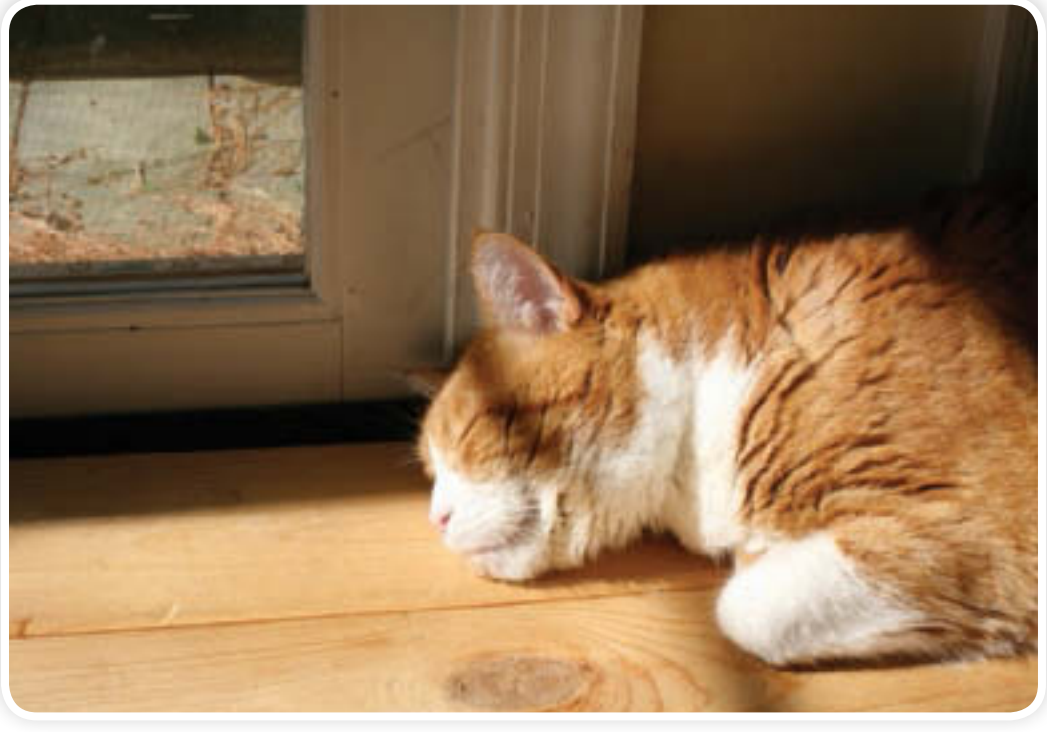
Energy Can Move and Change

Look at the napping cat in the picture. Not very energetic, is he? But there is energy in this scene that you might not notice at first.

The cat sleeps in a patch of sunlight. Light from the sun travels through space and transfers energy all the way to Earth. The sunlight pours through the screen door into the room where the cat is sleeping.

The light enters the room, and when it hits an object, the energy of light is changed to the energy of heat. So, the cat isn't moving, but energy that has moved across millions of miles is now in the room and warming the cat.

In later lessons, you will learn more about how energy can move from one place to another and change from one form to another.



Light and heat are forms of energy that allow this cat to take a cozy nap.